

9 Review

9.4 Final Review

Example 9.4.1 Write the number 3042_5 in base 10.

Example 9.4.2 Write 1024 in base 6.

Example 9.4.3 Add the following Mayan numerals together.

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Example 9.4.4 Let's remember how shift ciphers work.

ABCDEFGHIJKLMNOPQRSTUVWXYZ
ABCDEFGHIJKLMNOPQRSTUVWXYZ

Example 9.4.5 Encrypt the message “good luck on the final” using a transposition cipher, and secret key “lucky”.

Example 9.4.6 The following is an encrypted message using a transposition cipher with secret key “goalie”. What was the message?

ntfrin eafsfm leehat tgiaw

Example 9.4.7 Your turn! What does the key exchange algorithm look like with $p = 19$ and $g = 13$?

For these formulas, we have the following: P_0 is the principal. r is the interest rate. t is the number of time intervals (specified by problem) which have passed. A is the amount paid back. I is the interest.

Simple Interest	$A = P_0(1 + r)$
Simple Interest over Time	$A = P_0(1 + rt)$

For these formulas, we have the following: P_0 is the principal, or amount of a loan. r is the annual interest rate. t is the number of years which have passed. A is the total amount. k is the number of times interest is compounded per year. d is the amount we deposit (or withdraw) from the account regularly, or the payment amount on a loan.

Compound Interest	$A = P_0 \left(1 + \frac{r}{k}\right)^{kt}$
Savings Annuity	$A = d \frac{\left(\left(1 + \frac{r}{k}\right)^{kt} - 1\right)}{\frac{r}{k}}$
Payout Annuity/Loans	$P_0 = d \frac{\left(1 - \left(1 + \frac{r}{k}\right)^{-kt}\right)}{\frac{r}{k}}$

Example 9.4.8 How much do you need in a retirement account to withdraw \$80,000 a year for 30 years? Assume that the retirement account earns 5% interest.

Example 9.4.9 Paul wants to save up \$10,000 in 5 years. How much will he need to deposit a month in an account earning 4% to achieve this?

Example 9.4.10 If you open an account with 5% interest, how much will you need to deposit when the account opens in order to have \$20,000 in 20 years?

Example 9.4.11 Miles is going to take out a loan for a new car, if she can afford \$235 monthly payments, how big a loan can she take out if the loan earns 3% interest and is 5 years long?